BURHAN KARAHASAN

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EDUCATION

Koç University, Istanbul — M.Sc., Mechanical Engineering — CGPA 4.0/4.0 · 2024 – Present
Koç University, Istanbul — B.Sc., Mechanical Engineering *(Third Rank)* — 2018 – 2024
Double Major: B.Sc., Electrical & Electronics Engineering — CGPA 3.95/4.0 · Summa Cum Laude

RESEARCH EXPERIENCE

Koç University Medical Robotics Laboratory (Prof. Metin Sitti) — 2024 – Present
Graduate Researcher — *Istanbul, Türkiye*

- **X-ray image-guided closed-loop control of a soft magnetic millirobot.** Developed a platform that closes the loop between live X-ray imaging, robot localization and magnetic actuation; now extending it to a professional robotic manipulator.
- Designed a finite-state-machine controller in place of PID, with dedicated states for real-time stepwise navigation, occlusion removal and stall recovery; it gives stable, precise closed-loop navigation under delayed, discrete actuation.
- Adapted an existing neural-network detector from prior group work (C. Wang et al., *Nature Machine Intelligence*, 2026) for real-time robot localization in low-contrast X-ray frames.
- Video: youtu.be/_P9mQ2cJjDY
- **Ongoing: physics-based modeling and design of soft magnetic millirobots.** Developing models of untethered soft magnetic millirobots to design and control the robots I build; manuscript in preparation, details withheld for now.

Koç University Manufacturing and Automation Research Center (Prof. Ismail Lazoglu) — 2021 – 2024
Undergraduate Research Assistant — *Istanbul, Türkiye*

- Robot-guided surgery: surgical path optimization avoiding critical zones and brain vessels.
- Mechanical ventilation device: worked on the serial communication of several sensors & devices, optimized the system and improved the sampling rate of sensors 20 times.

PUBLICATIONS & PRESENTATIONS

Peer-reviewed journal article
Munam Arshad, Eda Guven, **Burhan Karahasan**, Ismail Lazoglu. "A novel real-time wireless sensor integration for enhancing positive pressure system operation in single limb passive vented circuit." *Biomedical Signal Processing and Control* 85 (2023).

Manuscript under review
Burhan Karahasan, Ismail Lazoglu, Ihsan Solaroglu. "Path optimization for robotic surgery avoiding functional zones" (full title withheld pending review). Submitted to *Computer Methods and Programs in Biomedicine*; under review, not yet accepted.

Conference presentation (poster)
Burhan Karahasan, Ömer Yiğit, Chunxiang Wang, Metin Sitti. "Image-Guided Small-Scale Magnetic Robot Navigation Under X-Ray." Poster, 7th Turkish Robotics Conference (national), Koç University, İstanbul, July 2026.

Manuscripts in preparation (not yet submitted)
Burhan Karahasan et al. Physics-based modeling of soft magnetic millirobots (title withheld).
Burhan Karahasan et al. Journal manuscript extending the X-ray navigation work above (title withheld).

TECHNICAL SKILLS

Programming	MATLAB, Python, Java/JavaFX, C/C++, VHDL, Visual Basic.
CAD	Siemens NX, SolidWorks, Fusion 360, Autodesk Inventor.
Simulation	Ansys Mechanical, Ansys Fluent, COMSOL Multiphysics.
Miscellaneous	Robotics, Machine Learning/Deep Learning, Computer Vision, Finite Element Analysis, Computer Aided Design/Manufacturing, Numerical Methods, Generative Design, LaTeX.
Languages	English (full professional; all degrees taught in English), Chinese (fundamental).

ACHIEVEMENTS

Elected Presidential Fellow of Koç University (2024).
Completed the track program in artificial intelligence.
Received Vehbi Koç Honor awards for 6 semesters.
DEVAK Scholarship recipient.
Full merit scholarship recipient for the undergraduate education in Koç University.

INDUSTRY EXPERIENCE

Ubicro, Prototyping and Testing Engineer — Aug. – Dec. 2022
DeltaV Space Technologies, GN&C Intern — Aug. – Sep. 2022
Turkish Aerospace Industries, Integrated Logistics Support Intern — June – July 2021

LEADERSHIP & TEACHING

- Team Leader, Koç University Autonomous Drone Team (KUADRONE), TÜBİTAK UAV Competition (2020 – 2021): mission planning, thrust calculations, circuit design.
- Teaching assistant and peer academic support: Calculus I, Statics, Dynamics, Dynamical Modeling & Control, Senior Design Project.

RELEVANT COURSES

Graduate — Medical Image Analysis, Numerical Modeling & Simulation, Computer-Based Simulation & Modeling, Optimization Models & Algorithms, Mechanics & Materials in Medicine, Biological Fluid Mechanics.
Undergraduate — Robotics, Introduction to Machine Learning, Computer Vision with Deep Learning, Linear Systems Theory.

OBJECTIVE

Robotics researcher with end-to-end experience across robot hardware, imaging, real-time control, and physics-based modeling of soft robots. I aim to pursue a PhD in the control and autonomy of small-scale and medical robots.

REFERENCES

Prof. Metin Sitti — M.Sc. advisor; President, Koç University — msitti@ku.edu.tr
Prof. Ismail Lazoglu — Undergraduate research advisor, Koç University — ilazoglu@ku.edu.tr
Prof. Alper Demir — Electrical and Electronics Engineering, Koç University — aldemir@ku.edu.tr